SeedHammer II — the pathological wallet

Backing up the constellation's **pathological example** from files: 11 keys, all four Bitcoin timelock kinds, and a hashlock. Host CLIs, the firmware GUI in the emulator, and the plates. No NFC anywhere.

Test material only — never put funds behind these keys.
The three masters are BIP-39's own published test vectors (`abandon…about`, `legal winner…yellow`, `letter advice…above`). They are public by construction.

The wallet

From `mnemonic-toolkit`'s Examples §5, "Custom degrading-miniscript wallet — **the pathological example**". A four-tier vault that deliberately mixes every timelock kind Bitcoin has:

Tier	Spend condition	Timelock	Keys
1	3-of-3 + secret word	`after(1000000)` — absolute **height**	@0 @1 @2
2	2-of-3 + secret word	`after(1893456000)` — absolute **time** (2030-01-01)	@3 @4 @5
3	both	`older(65535)` — relative **blocks** (~455 d)	@6 @7
4	any 1 of 3	`older(4255898)` — relative **time** (~365 d)	@8 @9 @10

The secret word is `opensessame`; the descriptor commits to `H = sha256(sha256(word))`, shared by tiers 1 and 2. Reusing a *hash* across tiers is fine — it is not a key. The 11 keys come from three masters at divergent account indices (`48'/0'/0..3'/2'`), so no key is reused.

This is the case that actually forces chunking.
The 12-key `multi(5,…)` policy tried first encodes to 13 bytes and one md1 string — a keyless BIP-388 template does not pay per key. This one carries two 32-byte hash literals and four timelock arguments, and comes to **182 data symbols against a single-string cap of 80**. It is the first policy in this work that genuinely needs a chunk set.

Toolchain

```
$ /scratch/code/shibboleth/descriptor-mnemonic/target/release/md --version
md 0.13.0
[exit 0]

$ /scratch/code/shibboleth/mnemonic-key/target/release/mk --version
mk 0.13.0
[exit 0]

$ /scratch/code/shibboleth/mnemonic-secret/target/release/ms --version
ms 0.16.0
[exit 0]

$ /scratch/code/shibboleth/mnemonic-engrave/target/release/me --version
me 0.7.0
[exit 0]

$ /scratch/code/shibboleth/mnemonic-engrave/target/release/me-preview --version
me-preview 0.7.0
[exit 0]
```

The input files

THE POLICY

inputs-pathological/wallet-policy.txt

```
wsh(or_i(and_v(v:after(1000000),and_v(v:sha256(a84dce40975727c398023cfbd50d5db3b9662375521d0f1ac62dbd829b9a08ad),multi(3,@0/<0;1>/*,@1/<0;1>/*,@2/<0;1>/*))),or_i(and_v(v:after(1893456000),and_v(v:sha256(a84dce40975727c398023cfbd50d5db3b9662375521d0f1ac62dbd829b9a08ad),multi(2,@3/<0;1>/*,@4/<0;1>/*,@5/<0;1>/*))),or_i(and_v(v:older(65535),multi(2,@6/<0;1>/*,@7/<0;1>/*)),and_v(v:older(4255898),multi(1,@8/<0;1>/*,@9/<0;1>/*,@10/<0;1>/*))))))
```

THE THREE MASTERS (SECRET — BIP-39 TEST VECTORS)

inputs-pathological/seeds/master-A.seed

```
abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon about
```

inputs-pathological/seeds/master-B.seed

```
legal winner thank year wave sausage worth useful legal winner thank yellow
```

inputs-pathological/seeds/master-C.seed

```
letter advice cage absurd amount doctor acoustic avoid letter advice cage above
```

THE ELEVEN KEYS

inputs-pathological/keys/key-00.xpub

```
# @0 — tier 1 (3-of-3, after height 1000000 + secret word)
# master A, origin [73c5da0a/48'/0'/0'/2']
xpub6DkFAXWQ2dHxq2vatrt9qyA3bXYU4ToWQwCHbf5XB2mSTexcHZCeKS1
VZYcPoBd5X8yVcbXFHJR9R8UCVpt82VX1VhR28mCyxUFL4r6KFrf
```

inputs-pathological/keys/key-01.xpub

```
# @1 — tier 1
# master A, origin [73c5da0a/48'/0'/1'/2']
xpub6DzhyrnFFYQ1HimDiM388xHnDiRPNdZJFBmmxge3Y1WWcHLtMJLfRuh
RHqnQCPbTj3fGKTuKFLHzzwpJkp5Dtc3UtLKZKaVZe1yqMBXd6Vk
```

inputs-pathological/keys/key-02.xpub

```
# @2 — tier 1
# master A, origin [73c5da0a/48'/0'/2'/2']
xpub6EGx8sPr9FxPPE1rbZazhqWwpMXA3Hf5DYKtZbL7c4BSddzmQktp96U
aTvecEkoCZysuaj79GMCFZYT1KKk7Ph2M3Kf5g8B82KZ8TZ9SKQR
```

inputs-pathological/keys/key-03.xpub

```
# @3 — tier 2 (2-of-3, after time 1893456000 + secret word)
# master A, origin [73c5da0a/48'/0'/3'/2']
xpub6E6Z3Ss5TXJYNJp4U1q3NZ3pCn82i7KXQAKUtNnzLJ3cCdchQeSdFvX
emizaHUF7wNwRQAB8mPdoZhGHLiv49cWPtCnoJY3Az3E8JKxH9Mq
```

inputs-pathological/keys/key-04.xpub

```
# @4 — tier 2
# master B, origin [b8688df1/48'/0'/0'/2']
xpub6FQya7zGhR92kacYsNnjreouvnHJMpXYsUXnW6NJJAJRCKsa26TzDy4
LdnGhEurr3d6y1J8PJ7EEMKQp74XTqYvmGJNogYXSKDszYHtF8mX
```

inputs-pathological/keys/key-05.xpub

```
# @5 — tier 2
# master B, origin [b8688df1/48'/0'/1'/2']
xpub6EAMBJLn1jiquajTsNRkZXU1oKnA4WJMNvcz4FRR4QmFKdfHxJVvfRL
oysWfcc16AMTR4CoMD8UNjvs9JtbsLeuLwpTczgq8zuuERnp8YZF
```

inputs-pathological/keys/key-06.xpub

```
# @6 — tier 3 (both, older 65535 blocks)
# master B, origin [b8688df1/48'/0'/2'/2']
xpub6EGceQV5wHrRemjYkJykRWQefTvsNTw1Lr7JUpQTgaaWz2P6Cg75o3L
ktGoyMPshKbCqrgy2RXxAmkqBXkbyqPe52CnwA1LUnyBkNUuMhRt
```

inputs-pathological/keys/key-07.xpub

```
# @7 — tier 3
# master B, origin [b8688df1/48'/0'/3'/2']
xpub6ELyn7moeEZ9hJ1HmSm6G5hAfgcCeTa71X9PzYJ39tHNUj93e9MaKb2
tjAFrMFb94NMeTG7MW8dwxuhhnPhn5swY6r5dxZH6cyiuPd25AQ5
```

inputs-pathological/keys/key-08.xpub

```
# @8 — tier 4 (any 1 of 3, older 4255898 = ~365d)
# master C, origin [28645006/48'/0'/0'/2']
xpub6DnEBNkSJKBYQmsbhS1sP9cNdtU5c9PLFGCjTJmxicxc13WB8zNNGQa
zabQpyFAGW5bV9tMko4uBxDxjUKL6dSAcx1tEbgEHtgSqyRsekh6
```

inputs-pathological/keys/key-09.xpub

```
# @9 — tier 4
# master C, origin [28645006/48'/0'/1'/2']
xpub6F6gx8ZP9R3R3eYsU2PeS5EPJ4jN7Wbt9uwyHNXLoaJxQjNT92FGAfC
NDjUDRhCHwzjfgDuqAZ7Gk9SugPRMa6A8PnzLVvnyEKBW9jHRGRp
```

inputs-pathological/keys/key-10.xpub

```
# @10 — tier 4
# master C, origin [28645006/48'/0'/2'/2']
xpub6Dh4twkBRkzxDSau8FdhVY16jARGnPtfdRUk6LbePrGUGok6xE8SvGt
aygCMLV7beB55YtzkR9dJoic9zYbYy3C7EKc2FGLkKoPT4mQUjPg
```

Host step 1 — it does not fit one string

```
$ cat /scratch/code/shibboleth/mnemonic-engrave/design/journeys/inputs-pathological/wallet-policy.txt
wsh(or_i(and_v(v:after(1000000),and_v(v:sha256(a84dce40975727c398023cfbd50d5db3b9662375521d0f1ac62dbd829b9a08ad),multi(3,@0/
<0;1>/*,@1/<0;1>/*,@2/<0;1>/*))),or_i(and_v(v:after(1893456000),and_v(v:sha256(a84dce40975727c398023cfbd50d5db3b9662375521d0
f1ac62dbd829b9a08ad),multi(2,@3/<0;1>/*,@4/<0;1>/*,@5/<0;1>/*))),or_i(and_v(v:older(65535),multi(2,@6/<0;1>/*,@7/<0;1>/*)),a
nd_v(v:older(4255898),multi(1,@8/<0;1>/*,@9/<0;1>/*,@10/<0;1>/*))))))
[exit 0]
```

```
$ /scratch/code/shibboleth/descriptor-mnemonic/target/release/md encode --group-size 0 wsh(or_i(and_v(v:after(1000000),and_v
(v:sha256(a84dce40975727c398023cfbd50d5db3b9662375521d0f1ac62dbd829b9a08ad),multi(3,@0/<0;1>/*,@1/<0;1>/*,@2/<0;1>/*))),or_i
(and_v(v:after(1893456000),and_v(v:sha256(a84dce40975727c398023cfbd50d5db3b9662375521d0f1ac62dbd829b9a08ad),multi(2,@3/<0;1
>/*,@4/<0;1>/*,@5/<0;1>/*))),or_i(and_v(v:older(65535),multi(2,@6/<0;1>/*,@7/<0;1>/*)),and_v(v:older(4255898),multi(1,@8/<0;
1>/*,@9/<0;1>/*,@10/<0;1>/*))))))
md: codec error: payload is 182 data symbols; the codex32 regular code caps single strings at 80 (use chunked encoding / --f
orce-chunked)
[exit 1]
```

Host step 2 — so it is chunked, with an explicit origin

Encoded without `--path`, this policy's card carries no origin and `md` says so: its `wsh(or_i(…))` wrapper has no canonical default derivation path, so `md decode` only PARTIAL-decodes (exit 4, VERIFY-ME) and `me bundle` rejects the set outright. Supplying the origin the warning asks for is what makes the rest of the chain work.

```
$ /scratch/code/shibboleth/descriptor-mnemonic/target/release/md encode --group-size 0 --force-chunked --path bip48 wsh(or_i
(and_v(v:after(1000000),and_v(v:sha256(a84dce40975727c398023cfbd50d5db3b9662375521d0f1ac62dbd829b9a08ad),multi(3,@0/<0;1>/*,
@1/<0;1>/*,@2/<0;1>/*))),or_i(and_v(v:after(1893456000),and_v(v:sha256(a84dce40975727c398023cfbd50d5db3b9662375521d0f1ac62db
d829b9a08ad),multi(2,@3/<0;1>/*,@4/<0;1>/*,@5/<0;1>/*))),or_i(and_v(v:older(65535),multi(2,@6/<0;1>/*,@7/<0;1>/*)),and_v(v:o
lder(4255898),multi(1,@8/<0;1>/*,@9/<0;1>/*,@10/<0;1>/*))))))
chunk-set-id: 0x829c6
md1fs2wxpqpzjtvyyy4qqxpx2v6mqq85yszv6a4pxuusyh2unu8xqz8naa2r2akwukvgm42gwsc92gdhnxq3mrt4
md1fs2wxpqwxkx9k7c9xu6pzk3ssspyefntdcdhkyqfntk5ymnjqjatj0sucqg70h4gdtkemjesdfjpw8z4kp2ay
md1fs2wxpq3rw4fp6rc6cckmmq5mngy26xpzx3t9xdwqqq8lluvzze6v6uqpq0pxscq2y6qqh8tw3xyv8tr5s
note: stdout is a keyless descriptor template (no keys)
[exit 0]
```

What `--path` does and does not say.

It records ONE shared origin — `bip48` and `m/48'/0'/0'/2'` produce a byte-identical chunk set here. These eleven keys actually sit at four account indices (`48'/0'/0..3'/2'`) across three masters, and each mk1 card carries its own true `origin_path`. So the md1's shared value is a default the key cards override; a restore that trusted the descriptor card alone would derive the wrong keys for @3–@10. Worth pinning with a restore test before this shape is recommended to anyone.

Host step 3 — the chunk set decodes back

Three cards, reassembled, give the same 11-key policy back — the round trip that makes the backup worth engraving.

```
$ /scratch/code/shibboleth/descriptor-mnemonic/target/release/md inspect md1fs2wxpqpzjtvyyy4qqxpx2v6mqq85yszv6a4pxuusyh2unu8
xqz8naa2r2akwukvgm42gwsc92gdhnxq3mrt4 md1fs2wxpqwxkx9k7c9xu6pzk3ssspyefntdcdhkyqfntk5ymnjqjatj0sucqg70h4gdtkemjesdfjpw8z4kp2
ay md1fs2wxpq3rw4fp6rc6cckmmq5mngy26xpzx3t9xdwqqq8lluvzze6v6uqpq0pxscq2y6qqh8tw3xyv8tr5s
template: wsh(or_i(and_v(v:after(1000000),and_v(v:sha256(a84dce40975727c398023cfbd50d5db3b9662375521d0f1ac62dbd829b9a08ad),m
ulti(3,@0/<0;1>/*,@1/<0;1>/*,@2/<0;1>/*))),or_i(and_v(v:after(1893456000),and_v(v:sha256(a84dce40975727c398023cfbd50d5db3b96
62375521d0f1ac62dbd829b9a08ad),multi(2,@3/<0;1>/*,@4/<0;1>/*,@5/<0;1>/*))),or_i(and_v(v:older(65535),multi(2,@6/<0;1>/*,@7/<
0;1>/*)),and_v(v:older(4255898),multi(1,@8/<0;1>/*,@9/<0;1>/*,@10/<0;1>/*))))))
n: 11
wallet-policy-mode: false
md1-encoding-id: 829c6ddb2f65a7f47078dd60d5a57ce7
wallet-descriptor-template-id: 5b48af35d4321a3ac18b43045e2523cc
origins:
  @0: m/48'/0'/0'/2'
  @1: m/48'/0'/0'/2'
  @2: m/48'/0'/0'/2'
  @3: m/48'/0'/0'/2'
  @4: m/48'/0'/0'/2'
  @5: m/48'/0'/0'/2'
  @6: m/48'/0'/0'/2'
  @7: m/48'/0'/0'/2'
```

```
  @8: m/48'/0'/0'/2'
… clipped; full text in out/transcript.txt
```

Obstacle 1 — `mk` cannot derive the policy stub from a chunked md1

Every mk1 key card must carry a 4-byte policy-id stub; `mk encode` refuses without one. The only automatic way to get it is `--from-md1`, and that path rejects a chunk:

```
$ /scratch/code/shibboleth/mnemonic-key/target/release/mk encode --xpub xpub6DkFAXWQ2dHxq2vatrt9qyA3bXYU4ToWQwCHbf5XB2mSTexc
HZCeKS1VZYcPoBd5X8yVcbXFHJR9R8UCVpt82VX1VhR28mCyxUFL4r6KFrf --origin-fingerprint 73c5da0a --origin-path m/48'/0'/0'/2' --fro
m-md1 md1fs2wxpqpzjtvyyy4qqxpx2v6mqq85yszv6a4pxuusyh2unu8xqz8naa2r2akwukvgm42gwsc92gdhnxq3mrt4 --group-size 0
error: md1 input rejected: wire-format version mismatch: got 9, expected 4
[exit 2]
```

`mk` vendors `md-codec 0.34.0`; the primary crate is at `0.42.0`. The "version 9" it will not parse is the chunked wire form.

WORKING AROUND IT — AND A SPEC/IMPLEMENTATION DIVERGENCE FOUND ON THE WAY

The stub is a property of the *descriptor*, not of the md1 string, so it is well defined for a chunked policy; `md inspect` prints the identities. But which one? `SPEC_mk_v0_1.md` §3.3 is explicit — "the top 4 bytes of the MD-encoded policy's **WalletPolicyId**" — and today's `md` prints a field by exactly that name. **That is not the one `mk` uses.** Measured on a single-string wallet where `--from-md1` does work:

```
wallet-descriptor-template-id: 726a666305756435b7c52c5b3fc69c41
wallet-policy-id:              f05e8a1c282f7740bbfd902a759b5577
policy_id_stubs (what mk embedded):  726a6663
```

The stub tracks the **template-id**, not the `wallet-policy-id`. Most likely a rename across versions — `mk` vendors md-codec 0.34.0 and the primary is 0.42.0, which now exposes both — but as it stands the spec sentence and the binary disagree about which identity a key card indexes.

```
$ bash -c /scratch/code/shibboleth/descriptor-mnemonic/target/release/md inspect md1fs2wxpqpzjtvyyy4qqxpx2v6mqq85yszv6a4pxuu
syh2unu8xqz8naa2r2akwukvgm42gwsc92gdhnxq3mrt4 md1fs2wxpqwxkx9k7c9xu6pzk3ssspyefntdcdhkyqfntk5ymnjqjatj0sucqg70h4gdtkemjesdfj
pw8z4kp2ay md1fs2wxpq3rw4fp6rc6cckmmq5mngy26xpzx3t9xdwqqq8lluvzze6v6uqpq0pxscq2y6qqh8tw3xyv8tr5s  2>/dev/null | grep -E 'pol
icy-id|template-id'
wallet-descriptor-template-id: 5b48af35d4321a3ac18b43045e2523cc
wallet-policy-id: f89e23f13c697ae62ef10328d71d7e24
wallet-policy-id-fingerprint: 0xf89e23f1
[exit 0]
```

Following the binary rather than the sentence, this wallet's stub is **`5b48af35`**, supplied with `--policy-id-stub`.

Host step 4 — the eleven key cards

```
note: 11 key cards -> 30 mk1 chunks (BIP-48 origins push most cards to 3)
$ wc -l /scratch/code/shibboleth/mnemonic-engrave/design/journeys/out/pathological/card-index.txt
30 /scratch/code/shibboleth/mnemonic-engrave/design/journeys/out/pathological/card-index.txt
[exit 0]

$ wc -l /scratch/code/shibboleth/mnemonic-engrave/design/journeys/out/pathological/mk-encode-raw.txt
30 /scratch/code/shibboleth/mnemonic-engrave/design/journeys/out/pathological/mk-encode-raw.txt
[exit 0]

$ /scratch/code/shibboleth/mnemonic-key/target/release/mk decode mk1qpd8cwpqqsq4kj90x4eutks2q5zg3vs7rnefw94m5rru59s2su80aw2q
4wgdpapgfl4pkhsdyytkwl5z8lphut2hvvpp5av5muuc0cmfrjw2 mk1qpd8cwpp806lhaeh6reknylagmwyjycf8044xtt9flsdlkvt6f6cthyl995lpm5zlrp6
yv6kc36tw
xpub:                xpub6DkFAXWQ2dHxq2vatrt9qyA3bXYU4ToWQwCHbf5XB2mSTexcHZCeKS1VZYcPoBd5X8yVcbXFHJR9R8UCVpt82VX1VhR28mCyxUF
L4r6KFrf
origin_fingerprint:  73c5da0a
origin_path:         48'/0'/0'/2'
policy_id_stubs:     5b48af35
chunks:              2 (long)
note: stdout is watch-only — public keys only, cannot spend
[exit 0]
```

Each key splits into **2 chunks**, so the eleven cards are 22 strings. Note the decode: the card carries the origin fingerprint and path, so the origins the descriptor card lacks are present in the bundle.

Host step 5 — the bundle validates, and names every plate

Every set verifies, the manifest emits, and the operator gets the number that matters before touching the machine.

```
$ /scratch/code/shibboleth/mnemonic-engrave/target/release/me bundle --in /scratch/code/shibboleth/mnemonic-engrave/design/j
ourneys/out/pathological/backup-strings.txt --preview /scratch/code/shibboleth/mnemonic-engrave/design/journeys/out/patholog
ical/plates --png --manifest /scratch/code/shibboleth/mnemonic-engrave/design/journeys/out/pathological/manifest.json
me: rendered plate 1 … (2–25 elided) … → /scratch/code/shibboleth/mnemonic-engrave/design/journeys/out/pathological/plates/p
late-1.png
me: wrote manifest to /scratch/code/shibboleth/mnemonic-engrave/design/journeys/out/pathological/manifest.json
me: backup needs 34 plates (33 public + ms1 on device):
  plate 1/34  md1 policy  → push via NFC & engrave
  plate 2/34  md1 policy  → push via NFC & engrave
  plate 3/34  md1 policy  → push via NFC & engrave
  plate 4/34  mk1 [b8688df1/48'/0'/1'/2'] chunk 1/3  → push via NFC & engrave
  plate 5/34  mk1 [b8688df1/48'/0'/1'/2'] chunk 2/3  → push via NFC & engrave
  plate 6/34  mk1 [b8688df1/48'/0'/1'/2'] chunk 3/3  → push via NFC & engrave
  plate 7/34  mk1 [73c5da0a/48'/0'/3'/2'] chunk 1/3  → push via NFC & engrave
  plate 8/34  mk1 [73c5da0a/48'/0'/3'/2'] chunk 2/3  → push via NFC & engrave
  plate 9/34  mk1 [73c5da0a/48'/0'/3'/2'] chunk 3/3  → push via NFC & engrave
  plate 10/34  mk1 [73c5da0a/48'/0'/2'/2'] chunk 1/3  → push via NFC & engrave
  plate 11/34  mk1 [73c5da0a/48'/0'/2'/2'] chunk 2/3  → push via NFC & engrave
  plate 12/34  mk1 [73c5da0a/48'/0'/2'/2'] chunk 3/3  → push via NFC & engrave
  plate 13/34  mk1 [73c5da0a/48'/0'/0'/2'] chunk 1/2  → push via NFC & engrave
  plate 14/34  mk1 [73c5da0a/48'/0'/0'/2'] chunk 2/2  → push via NFC & engrave
  plate 15/34  mk1 [28645006/48'/0'/0'/2'] chunk 1/2  → push via NFC & engrave
  plate 16/34  mk1 [28645006/48'/0'/0'/2'] chunk 2/2  → push via NFC & engrave
  plate 17/34  mk1 [28645006/48'/0'/2'/2'] chunk 1/3  → push via NFC & engrave
  plate 18/34  mk1 [28645006/48'/0'/2'/2'] chunk 2/3  → push via NFC & engrave
  plate 19/34  mk1 [28645006/48'/0'/2'/2'] chunk 3/3  → push via NFC & engrave
  plate 20/34  mk1 [73c5da0a/48'/0'/1'/2'] chunk 1/3  → push via NFC & engrave
  plate 21/34  mk1 [73c5da0a/48'/0'/1'/2'] chunk 2/3  → push via NFC & engrave
  plate 22/34  mk1 [73c5da0a/48'/0'/1'/2'] chunk 3/3  → push via NFC & engrave
  plate 23/34  mk1 [b8688df1/48'/0'/2'/2'] chunk 1/3  → push via NFC & engrave
  plate 24/34  mk1 [b8688df1/48'/0'/2'/2'] chunk 2/3  → push via NFC & engrave
  plate 25/34  mk1 [b8688df1/48'/0'/2'/2'] chunk 3/3  → push via NFC & engrave
  plate 26/34  mk1 [b8688df1/48'/0'/3'/2'] chunk 1/3  → push via NFC & engrave
  plate 27/34  mk1 [b8688df1/48'/0'/3'/2'] chunk 2/3  → push via NFC & engrave
  plate 28/34  mk1 [b8688df1/48'/0'/3'/2'] chunk 3/3  → push via NFC & engrave
  plate 29/34  mk1 [b8688df1/48'/0'/0'/2'] chunk 1/2  → push via NFC & engrave
  plate 30/34  mk1 [b8688df1/48'/0'/0'/2'] chunk 2/2  → push via NFC & engrave
  plate 31/34  mk1 [28645006/48'/0'/1'/2'] chunk 1/3  → push via NFC & engrave
  plate 32/34  mk1 [28645006/48'/0'/1'/2'] chunk 2/3  → push via NFC & engrave
  plate 33/34  mk1 [28645006/48'/0'/1'/2'] chunk 3/3  → push via NFC & engrave
  plate 34/34  ms1 secret  → TYPE ON DEVICE (New > Input Seed > CODEX32); never via this tool
[exit 0]
```

Host step 6 — the seed, and the refusal that still holds

The addresses — the check this wallet could never do

Until the eleven keys were re-derived at BIP-48's four-level origin, `md` refused every one of them (`expected depth 4 for this script context, got 3`) and **no address could be derived for this wallet by any tool**. That is why no earlier version of this document showed one. These are the structural half's counterpart: proof the bytes mean the wallet that was intended.

```
Item 5: the eleven keys used to sit at BIP-84's three-level single-sig
account path, while wsh(<miniscript>) is a MultiSig script context that
requires a depth-FOUR xpub. md refused every one of them, so NO ADDRESS
could be derived for this wallet by any tool -- which is why no journey
ever showed one. Re-derived at m/48'/0'/N'/2', they compose.

This is the FUNCTIONAL half of a round trip: the structural half proves
the bytes survived, and this proves they mean the wallet we intended.
Compare these against your coordinator before engraving anything.

$ /scratch/code/shibboleth/descriptor-mnemonic/target/release/md address md1fatzr2spzjtvyyy4qqxpx2v6mqq85yszv6a4pxuusyh2unu8
xqz8naa2r2akwukvgm42gwsr8ztkypux5za md1fatzr2s2rc6cckmmq5mngy26xzzqyn9xddhpk7cspxdw6snwwgzt4wf7rnqpre774p4wmxtyldgm8whvtd8 m
d1fatzr2s4evc3h25sapudvvtdas2de5z9drq3rg4jnxhqqqrll7xppvaxdwqqs8sngvq9zvd5z6s7pygwfk3 md1fatzr2su9k6zqhwsv0jskp2rsal4egz4ep5
859p875x67p5s3wem7sgluxl3d2a3syx3m74wyswrxcvpc54 md1fatzr23zlhaeh6reknylagmwyjycf8044xtt9flsdlkvt6f6cthyl9yvevrz28llsnr4qy50
0rkxyfe6hw0 md1fatzr23tkxgeal5wa4wy86389z3gq870g9n6rjwuq7cmsyjgffgkpqmffa5k8pmzr2lxjzdetr8v52fs2cv md1fatzr23n04upy3z0n8ek4a
j5kk9satjt7uv4am7jn7q2pvhex36qvgp5csmqq29aj78uss09z0mledqds8p md1fatzr23m7j8mxp87s87gwfzjsys8njyze0uj4852ajv5x7fyu88u23mx7vw
v6azw8d59kw70a2lmlks0w95 md1fatzr2jx88nfhcc7nza0ynr9p5d3exan7jvup3kx8742q6d662qzwwx3qg6rmwy2r0upegmfdvte0lgjxay md1fatzr2jdk
x6k89kfkaz2az67ajl8hyy967je7sks39z80kl7quz6axuvpffejsahhtp2x7gczf3clc26muz md1fatzr2j3w82jq5zap9302ynhp37ggd6z5u4emag0zr8gh9
upnjugr26jfvunvs35jvgdjkvf8nxwprt58vw md1fatzr2jmwxeza6dduqqnvtpl3vsgcj7vs58sl2hmlrnysxhs57y20d0sn3047m4jnr9cz5e8jumuwpep9lg
md1fatzr2n86g9y9evngtkrskaxskvp4qchq5qpv0pez9rarvlx32c8627y2x4unucwuan7ngdj6m8a4jm3a38 md1fatzr2ntm5awv08c2v0vktw5t6qu9g4xkm
v2dywrwkudljyuk9dam643wu0vf30ddl5vwq28tgapq6km0dg md1fatzr2n3g59kdh7306na4nhq32cd5w9c2uzlhxryrtzdu2qf455xvp5uc0ktmeyw4h7cajy
9dzdt2xm663g md1fatzr2nuqhu772hljrwlhzlncxfd22zx672chhdu5ajtemtchqtljnzwtnna6j74phutvjl0dhs8ws22759 md1fatzr25q9z7s576y0hwv6
a92khwlwfmwulc9zrqupz2zll50ju3dcmghtxtfv0y025ltk2rlu3mjgf4q49y md1fatzr25fnqlk8ycusz0quhfev0a3hqdghjmksz3ry92d3gv4ejtmu9f0zx
f3clxvtlnnvxgaz09mvs08va3 md1fatzr25hgcj5r8x9t5j54jfadzzykqz37sefnc0rhk5zchwu9cfljgk8tg3nlnyr6nv2e7vtf5ttahr4hn4 md1fatzr25e
fq2e50d8mg50ts8t8mquqwc90uynctuc4sv9py5lug68qthk65sax8thnpjetx05qdw2xnz9759 md1fatzr24xrn6wz3pd0pwa4xqjj8e7htel64394h6gme2g4
flvkpvqrj3us7k4x9dcp2a2uz2f0tdhlnxyd7a md1fatzr24w75nz845y9up7y276lt29ywu3usza3aqrp2qarn88u9dtnzuy --chain 0 --count 3
bc1qkuknuy6dsm0fq44cyyhzqy9wl3ex2n6ed39zxhx867l9wlh4yhlsejms64
bc1q6k2emh2epd57p5qfwswnahtphjlzsuc5x92c7zphr2lnnckzrpzqjdaraf
bc1q97cvjd2wj75hu62w4at6w4ay0mz3janslla3y3zrml4lglymm38qfh70pa
note: stdout is watch-only — public keys only, cannot spend
[exit 0]

$ /scratch/code/shibboleth/descriptor-mnemonic/target/release/md address md1fatzr2spzjtvyyy4qqxpx2v6mqq85yszv6a4pxuusyh2unu8
xqz8naa2r2akwukvgm42gwsr8ztkypux5za md1fatzr2s2rc6cckmmq5mngy26xzzqyn9xddhpk7cspxdw6snwwgzt4wf7rnqpre774p4wmxtyldgm8whvtd8 m
d1fatzr2s4evc3h25sapudvvtdas2de5z9drq3rg4jnxhqqqrll7xppvaxdwqqs8sngvq9zvd5z6s7pygwfk3 md1fatzr2su9k6zqhwsv0jskp2rsal4egz4ep5
859p875x67p5s3wem7sgluxl3d2a3syx3m74wyswrxcvpc54 md1fatzr23zlhaeh6reknylagmwyjycf8044xtt9flsdlkvt6f6cthyl9yvevrz28llsnr4qy50
0rkxyfe6hw0 md1fatzr23tkxgeal5wa4wy86389z3gq870g9n6rjwuq7cmsyjgffgkpqmffa5k8pmzr2lxjzdetr8v52fs2cv md1fatzr23n04upy3z0n8ek4a
j5kk9satjt7uv4am7jn7q2pvhex36qvgp5csmqq29aj78uss09z0mledqds8p md1fatzr23m7j8mxp87s87gwfzjsys8njyze0uj4852ajv5x7fyu88u23mx7vw
v6azw8d59kw70a2lmlks0w95 md1fatzr2jx88nfhcc7nza0ynr9p5d3exan7jvup3kx8742q6d662qzwwx3qg6rmwy2r0upegmfdvte0lgjxay md1fatzr2jdk
x6k89kfkaz2az67ajl8hyy967je7sks39z80kl7quz6axuvpffejsahhtp2x7gczf3clc26muz md1fatzr2j3w82jq5zap9302ynhp37ggd6z5u4emag0zr8gh9
upnjugr26jfvunvs35jvgdjkvf8nxwprt58vw md1fatzr2jmwxeza6dduqqnvtpl3vsgcj7vs58sl2hmlrnysxhs57y20d0sn3047m4jnr9cz5e8jumuwpep9lg
md1fatzr2n86g9y9evngtkrskaxskvp4qchq5qpv0pez9rarvlx32c8627y2x4unucwuan7ngdj6m8a4jm3a38 md1fatzr2ntm5awv08c2v0vktw5t6qu9g4xkm
v2dywrwkudljyuk9dam643wu0vf30ddl5vwq28tgapq6km0dg md1fatzr2n3g59kdh7306na4nhq32cd5w9c2uzlhxryrtzdu2qf455xvp5uc0ktmeyw4h7cajy
9dzdt2xm663g md1fatzr2nuqhu772hljrwlhzlncxfd22zx672chhdu5ajtemtchqtljnzwtnna6j74phutvjl0dhs8ws22759 md1fatzr25q9z7s576y0hwv6
a92khwlwfmwulc9zrqupz2zll50ju3dcmghtxtfv0y025ltk2rlu3mjgf4q49y md1fatzr25fnqlk8ycusz0quhfev0a3hqdghjmksz3ry92d3gv4ejtmu9f0zx
f3clxvtlnnvxgaz09mvs08va3 md1fatzr25hgcj5r8x9t5j54jfadzzykqz37sefnc0rhk5zchwu9cfljgk8tg3nlnyr6nv2e7vtf5ttahr4hn4 md1fatzr25e
fq2e50d8mg50ts8t8mquqwc90uynctuc4sv9py5lug68qthk65sax8thnpjetx05qdw2xnz9759 md1fatzr24xrn6wz3pd0pwa4xqjj8e7htel64394h6gme2g4
flvkpvqrj3us7k4x9dcp2a2uz2f0tdhlnxyd7a md1fatzr24w75nz845y9up7y276lt29ywu3usza3aqrp2qarn88u9dtnzuy --chain 1 --count 3
bc1qxtlrmq23lczrtx8l9x3nsfp3u5h7vt9zjmy94m6srl354ps9amus2q56lx
bc1q6keuktp5nwv0mtr593gj8p4yvpvt5gl4hwhzyxzuyu66mtj7leaquyk8qt
bc1qcyn83peqe3n6flckqwf3fvr2g0y2z8hplfxfm8nfecazxas4ux2sjjwuld
note: stdout is watch-only — public keys only, cannot spend
[exit 0]
```

```
$ cat /scratch/code/shibboleth/mnemonic-engrave/design/journeys/inputs-pathological/seeds/master-A.seed
abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon about
[exit 0]

$ /scratch/code/shibboleth/mnemonic-secret/target/release/ms encode --phrase abandon abandon abandon abandon abandon abandon
abandon abandon abandon abandon abandon about
ms10e ntrsq qqqqq qqqqq qqqqq qqqqq qqqqq qqcj9 sxraq 34v7f
word count: 12
language: english (BIP-39 checksum valid)
passphrase: not stored in ms1 (record separately if used)
warning: stdout carries private key material (can spend) — redirect or encrypt (e.g. '> file.txt' or '| age -e ...')
[exit 0]

$ /scratch/code/shibboleth/mnemonic-engrave/target/release/me --in /dev/fd/63 --hex
```

```
me: refusing to emit ms1 over NFC: ms1 is secret seed entropy and must never be transmitted by radio (RF eavesdropping ris
k). Enter it by hand on the device: New > Input Seed > CODEX32.
[exit 3]
```

The 25 public plates

Three descriptor chunks, then eleven key cards at two chunks each.

md1fs2wxpqpzjtvyyy4qqxpx2v
6mqq85yszv6a4pxuusyh2unu8x
qz8naa2r2akwukvgm4
2gwsc92gdhnxq3mrt4

plate 1 — md1 policy, chunk 1/3

md1fs2wxpqwxkx9k7c9xu6pzk3
ssspyefntdcdhkyqfntk5ymnjq
jatj0sucqg70h4gdtk
emjesdfjpw8z4kp2ay

plate 2 — md1 policy, chunk 2/3

md1fs2wxpq3rw4fp6rc6cckmmq
5mngy26xpzx3t9xdwqqq8lluvz
ze6v6uqpq0pxscq2y6
qqh8tw3xyv8tr5s

plate 3 — md1 policy, chunk 3/3

mk1qpqs7nzqqsq4kj90xkux3r0
3lcztpqyqsqygpqyqsqygrqyqs
qyg9qyqsqyqfz9jr
e2hecfftal3ejgrt
c20z98khcfck4vmj
fu50256f66n

plate 4 — @5 [b8688df1/48'/0'/1'/2'] chunk 1/3

mk1qpqs7nzpa0kav5cewq405s2
gtjexshv8pd6dpvcr2p3wpgqzc
7rjy286xe7dz4s05
4ug5dte8esaem8ax
3a6whx8nu9qg6sqc
85yf3uknf8e

plate 5 — @5 [b8688df1/48'/0'/1'/2'] chunk 2/3

mk1qpqs7nzz8566kpy4nftsszz
avq

plate 6 — @5 [b8688df1/48'/0'/1'/2'] chunk 3/3

mk1qpz0dqzqqsq4kj90x4eutks
2lcztpqyqsqygpqyqsqyg8qyqs
qyg9qyqsqyqfz9jr
exftdke6vt4ujvv5
x3k8ymk06fnswq4g
gh4rfvxqje5

plate 7 — @3 [73c5da0a/48'/0'/3'/2'] chunk 1/3

mk1qpz0dqzprrvv0a25p5m455q
yuudzq358kug5xlcrjjmrdtrjm
ymw39w3d0we0nmjz
za0fvlgtggj3rhm0
lqwpdwnwxq3xxzyn
sch9nz058e4

plate 8 — @3 [73c5da0a/48'/0'/3'/2'] chunk 2/3

mk1qpz0dqzzmg9ughf4wwykkyu
sha

plate 9 — @3 [73c5da0a/48'/0'/3'/2'] chunk 3/3

mk1qp8lruzqqsq4kj90x4eutks
2lcztpqyqsqygpqyqsqyg9qyqs
qyg9qyqsqyqfz9jr
ej0n5eghh0620cpg
9jly68gp3qxjpk5m
drj8ws2ulea

plate 10 — @2 [73c5da0a/48'/0'/2'/2'] chunk 1/3

mk1qp8lruzp3pkqq5tm9u0eppl
franqnlgrly8y3fgzgreezpvh7
f2n69wex2r0yjwrn
79gan0x8xdw38rk6
zm8cu7dxlrr73quu
qrkdr97f4xs

plate 11 — @2 [73c5da0a/48'/0'/2'/2'] chunk 2/3

mk1qp8lruzzcl8xvmvu2uxsczf
rum

plate 12 — @2 [73c5da0a/48'/0'/2'/2'] chunk 3/3

The 25 public plates, continued

mk1qpd8cwpqqsq4kj90x4eutks
2q5zg3vs7rnefw94m5rru59s2s
u80aw2q4wgdpapgf
l4pkhsdyytkwl5z8
lphut2hvvpp5av5m
uuc0cmfrjw2

plate 13 — @0 [73c5da0a/48'/0'/0'/2'] chunk 1/2

mk1qpd8cwpp806lhaeh6reknyl
agmwyjycf8044xtt9flsdlkvt6
f6cthyl995lpm5zlrp
6yv6kc36tw

plate 14 — @0 [73c5da0a/48'/0'/0'/2'] chunk 2/2

mk1qp073xpqqsq4kj90x55xg5q
xq5zg3vs7yxvkgrq39p0l68ewg
kud5t4n95k8j8420
4m9xvr7cunrjqfur
ja893lkxup4zlyyf
7spk75exfku

plate 15 — @8 [28645006/48'/0'/0'/2'] chunk 1/2

mk1qp073xpp09hdq9zxg25mzse
tnyhhc2j7yvnr37vchl8xc05vf
2pnnz46ffl6ymp2sa0
x397e56ypk

plate 16 — @8 [28645006/48'/0'/0'/2'] chunk 2/2

mk1qpjn45zqqsq4kj90x55xg5q
xlcztpqyqsqygpqyqsqyg9qyqs
qyg9qyqsqyqfz9jr
c2ha3f8aucvk2eu8
85u9zz67zam29pfm
mh35mcqwh32

plate 17 — @10 [28645006/48'/0'/2'/2'] chunk 1/3

mk1qpjn45zpxqjj8e7htel6439
4h6gme2g4flvkpvqrj3us7k4x9
dcp2a2urh4yc3adp
p0q03zhkh663frhy
0yqhv0gqc2egr4uv
xrm90z6cezn

plate 18 — @10 [28645006/48'/0'/2'/2'] chunk 2/3

mk1qpjn45zzad9jx7j942fx7se
wtu

plate 19 — @10 [28645006/48'/0'/2'/2'] chunk 3/3

mk1qp4dj9zqqsq4kj90x4eutks
2lcztpqyqsqygpqyqsqygrqyqs
qyg9qyqsqyqfz9jr
cld706hn9svfgll7
zvw5qnkxgea7y6pq
cgj2njpw0xx

plate 20 — @1 [73c5da0a/48'/0'/1'/2'] chunk 1/3

mk1qp4dj9zp68w6hzragnj3g5q
rl85zeape8wq0vdczfyy55tqsd
5576trsa3p40nfpd
7hsyjyf7vlx6hk2j
6ckr4wf0m36k7q09
20s9wqfx6hj

plate 21 — @1 [73c5da0a/48'/0'/1'/2'] chunk 2/3

mk1qp4dj9zzv308jhm5uzl5tlx
r6z

plate 22 — @1 [73c5da0a/48'/0'/1'/2'] chunk 3/3

mk1qp69p8zqqsq4kj90xkux3r0
3lcztpqyqsqygpqyqsqyg9qyqs
qyg9qyqsqyqfz9jr
ejrqpep8kt9h29aq
wz52ntdk9xjxstgy
ahmpt9894kj

plate 23 — @6 [b8688df1/48'/0'/2'/2'] chunk 1/3

mk1qp69p8zpsm4hr0u3893t0w7
4vthrmzvtmt0arrsz3gtvm0azl
48mt8wpz4smguts4
c9lwvxgxkymc5qnt
fgvcrfeslvazac9u
d8l8f472kz7

plate 24 — @6 [b8688df1/48'/0'/2'/2'] chunk 2/3

mk1qp69p8zzgfc7q43r84c4xca
t7f

plate 25 — @6 [b8688df1/48'/0'/2'/2'] chunk 3/3

mk1qpm4utzqqsq4kj90xkux3r0
3lcztpqyqsqygpqyqsqyg8qyqs
qyg9qyqsqyqfz9jr
ehxe4gmhjgat0a3m
xqt7009tlepkycdq
afqs2h9m2cf

plate 26 — @7 [b8688df1/48'/0'/3'/2'] chunk 1/3

mk1qpm4utzphachu7pjt2js3kh
jk9am098vj7w679czlu5cnjuul
w5h4gdlzmys29apf
a5glwue4624dwa7u
nkaels2yxpa6tycg
69v68twqw68

plate 27 — @7 [b8688df1/48'/0'/3'/2'] chunk 2/3

mk1qpm4utzz0zufgzze827cxpn
pdp

plate 28 — @7 [b8688df1/48'/0'/3'/2'] chunk 3/3

mk1qpmm93pqqsq4kj90xkux3r0
3q5zg3vs7llvu2xd8x2rk7av9g
mew82jq5zap9302y
nhp37ggd6z5u4ema
g0zr8gh9upnj5stu
xqtzdn4uxkd

plate 29 — @4 [b8688df1/48'/0'/0'/2'] chunk 1/2

mk1qpmm93ppwyp4dfykwfkgg6f
xyxetdcmythf4hsqzd3v879jpr
ztejzs7rl0vk8mlv7l
zva7w6nage

plate 30 — @4 [b8688df1/48'/0'/0'/2'] chunk 2/2

mk1qpuqa9zqqsq4kj90x55xg5q
xlcztpqyqsqygpqyqsqygrqyqs
qyg9qyqsqyqfz9jr
m20f24atyn66yyfv
q9rapjn8s78wrfec
27zfdk7lqv9

plate 31 — @9 [28645006/48'/0'/1'/2'] chunk 1/3

mk1qpuqa9zpk5zchwu9cfljgk8
tg3nlnyr6nv2e72gzkdrmf769r
6up6e7c8qrkptlpy
7zlx9vrpgf98lzx3
czaak4y8f37qz2nd
cmfw7e7qp9u

plate 32 — @9 [28645006/48'/0'/1'/2'] chunk 2/3

mk1qpuqa9zz8ffxj3xnc8vz05g
hwr

plate 33 — @9 [28645006/48'/0'/1'/2'] chunk 3/3

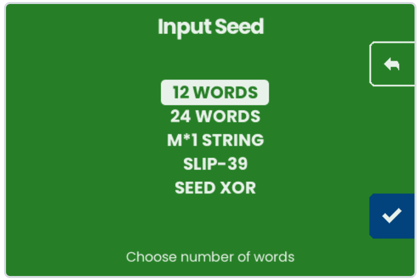
Plus three seed plates that are not here.

Each master is engraved from words typed on the machine. No host tool will render, transmit or preview them — so the real total is **28 plates**.

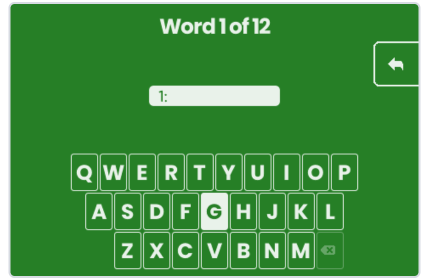
On the machine — the seed, typed

No NFC in this journey. The seed goes in the way the machine's own checklist says it must: by hand, on the device.

Boot: Backup Wallet



Input Seed — 12 words



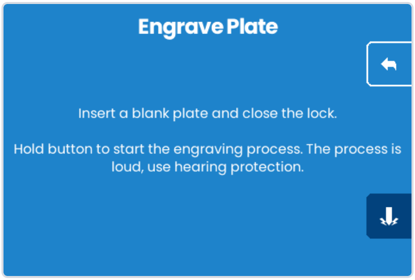
Word 1 of 12

Three letters is enough: “1 match”, auto-completed

All 12 accepted — the BIP-39 checksum holds

Optional BIP-39 passphrase, skipped

On the machine — the seed plate



Insert a blank plate; hold to start

Cutting

Still cutting — a seed plate is the long one

The plate overlay, live

The whole planned layout is drawn in **grey** the moment the engrave begins, from the same `PlanEngraving` call the firmware makes. What the driver has actually stepped is filled in **black**, decoded from the real step stream. The red circle is the head. The title is the master fingerprint `73C5DA0A` — the same one the descriptor's origins name.

head 25.9, 41.8 mm — 18% of the cut (34040352 steps)

head 24.1, 64.4 mm — 41% of the cut (77422644 steps)

head 56.7, 44.0 mm — 71% of the cut (134197290 steps)

head 59.4, 32.0 mm — 93% of the cut (177152436 steps)

What this run turned up

1. `mk encode --from-md1` cannot read a chunked md1.
`wire-format version mismatch: got 9, expected 4`, exit 2 — and a stub is mandatory, so for any policy large enough to need chunking the documented route to a key card does not work. `mk` vendors `md-codec 0.34.0` against the primary's `0.42.0`; version 9 is the chunked wire form its copy predates. The provenance pin is what should have caught this. Workaround: read the identity out of `md inspect` and pass `--policy-id-stub` by hand — which nothing tells an operator to do, and which requires knowing the next finding.

2. The spec and the binary disagree about which identity the stub indexes.
`SPEC_mk_v0_1.md` §3.3 says the stub is the top 4 bytes of the **WalletPolicyId**, and `md` prints a field of exactly that name. Measured, `mk` embeds the top 4 bytes of the **wallet-descriptor-template-id** instead — `726a6663` where the `wallet-policy-id` began `f05e8a1c`. Probably a rename that landed in md-codec after 0.34.0, but the sentence and the code now name different things, and a stub is the index a recovering operator uses to tell wallets apart.

3. `--path` is required here, and it flattens.
Without it the descriptor card carries no origin, `md decode` PARTIAL-decodes (exit 4) and `me bundle` refuses the set. With it, everything validates — but it records a single shared origin while these eleven keys sit at four account indices. The true paths survive on the mk1 cards; the descriptor card alone would restore @3–@10 wrongly. This is the one finding here that is a *design* question rather than a defect: a restore test should pin which source wins.

4. `md encode` has no per-key origin flag.
`--key` takes a bare xpub and rejects an origin-annotated one (`base58check decode`), and `--fingerprint` carries no path. So a divergent-origin wallet cannot state its origins in the descriptor card at all — only the flattened form of finding 3 is expressible.

5. Carried over from the first journey:
presenting an NFC tag to a gathering flow freezes the emulator (filed as **F-126**). This journey avoids NFC entirely, so it is not exercised here.

Corrected from the first draft of this document.
It reported `me bundle`'s refusal as a tool defect and rendered the plates by calling the sidecar directly. That was wrong: the refusal was the consequence of omitting `--path`, which `md` had warned about at encode time. The plates and checklist here come from `me bundle --preview`.

Reproducing this document

```
cd <this directory>
bash transcript_pathological.sh > transcript_pathological.txt 2>&1   # every CLI block, refusals included
python3 capture_pathological.py                                       # the 13 screenshots, from the emulator
python3 build_pdf_pathological.py                                     # this document, as a PDF
```

Emulator frames were captured by POSTing `canvas.toDataURL()` to a local receiver, so each screenshot is the device framebuffer exactly. Plate overlays are the page's own SVG, rasterised in the browser that drew them. Both are produced by `capture_pathological.py`, which drives `cmd/emu/shots_pathological.js` — the capture used to be console code in a session that no longer exists, which is why this document could not be rebuilt (F-210).